TRANSFORMER SUMMARIZER — COMPLETE EVALUATION PIPELINE

Article → 3 Models → 4 Evaluation Modules → Metrics Dashboard | CNN/DailyMail · 280k Articles · spaCy NER · Logistic Regression · Random Forest · ROUGE

STAGE 1 · INPUT

[DOC] INPUT ARTICLE (CNN/DailyMail Test Set)

SAMPLE ARTICLE TEXT:

"Scientists discovered a massive black hole located near Earth, raising new questions about our understanding of galactic formation. The team, led by Dr. Sarah Chen at MIT, used the James Webb Space Telescope to detect the object, which has a mass 10 billion times that of our Sun. Researchers believe it formed during the early universe..."

CORPUS	ARTICLES	SPLIT	AVG LENGTH	TOKENIZER
CNN / DailyMail	280,327 total	train / val / test	~780 tokens	WordPiece / BPE

[REF] REFERENCE SUMMARY

Human-written gold standard (Highlight field in dataset)

★ Used ONLY for intrinsic evaluation (ROUGE + NER)

STAGE 2 · MODELS

extractive

abstractive (sequential)

abstractive (parallel)

(1) TF-IDF EXTRACTIVE BASELINE

APPROACH: Extractive — copies full sentences from the original article.

STEP 1 Tokenize article → split sentences

STEP 2 Compute TF for each word:

$TF(w,s) = count(w \text{ in } s) / |s|$

STEP 3 Compute IDF over 280k articles:

$IDF(w) = \log(N / df(w))$

N=280327, df=doc freq of word

STEP 4 TF-IDF(w,s) = TF x IDF

STEP 5 Score each sentence:

score(s) = sum TF-IDF(w,s)

(content words only, skip stop)

STEP 6 Rank sentences -> pick top-K

K=2 for this project

EXAMPLE: "cat sat on mat" — 4 words

TF("cat",s) = 1/4 = 0.25

IDF("cat") = log(280327/1200)

= log(233.6) = 5.45

TF-IDF = 0.25 x 5.45 = 1.36

LIMIT: Cannot paraphrase — bounded by lexical overlap with reference.

ROUGE-1: 17.0% ROUGE-2: 6.1% ROUGE-L: 9.2%

High precision, low recall.

(2) BiLSTM + BAHDANAU ATTENTION

APPROACH: Abstractive — generates new words via encoder-decoder architecture.

ENCODER: Bidirectional LSTM

-> forward LSTM h->

<- backward LSTM h<-

concat: h_i = [h->_i ; h<-_i]

captures left + right context

DECODER: Unidirectional LSTM (sequential)

hidden state s_t updated at each

decoding step t = 1,2,...,T

BAHDANAU: Single-head additive attention

$e(t,i) = v^*\tanh(W_s s_t + W_h h_i)$

$a(t,i) = \mathrm{softmax}(e(t,i))$

$context\ c_t = \sum a(t,i) h_i$

one attention pattern per step

WEAKNESS: Sequential -> long-range decay

Cannot parallelize across time.

Single attention head misses

syntax/semantics simultaneously.

VS TRANS: Transformer attends all positions

at once — no decay, 4 heads.

ROUGE-1: ~20% ROUGE-2: ~8% ROUGE-L: ~14% (approx)

(3) TRANSFORMER (Seq2Seq)

APPROACH: Abstractive — parallel attention

No recurrence — pure attention.

ENCODER: 6 layers x 4-head self-attention

d_model=512, d_k=d_v=128

Q=X*W_Q, K=X*W_K, V=X*W_V

MH-ATTN: Attn(Q,K,V)=softmax(QKt/sqrt(d_k))*V

head_i: attends d_k=128 dims

concat all heads -> linear W_O

4 heads learn simultaneously:

Head 1 -> syntactic structure

Head 2 -> semantic similarity

Head 3 -> co-reference links

Head 4 -> local n-gram context

DECODER: 6 layers, masked self-attn +

cross-attention to encoder out

PARALLEL: All positions computed at once

-> no sequential decay

-> faster training (GPU-friendly)

VOCAB: BPE vocabulary, generates tokens

-> subword units -> paraphrase

discovered->found, massive->huge

ROUGE-1: 24.5% ROUGE-2: 9.8% ROUGE-L: 19.9% * BEST

STAGE 3 · SUMMARIES

"Scientists discovered a massive black hole
located near Earth, raising new questions."
"The team, led by Dr. Sarah Chen at MIT,
used James Webb Space Telescope to detect."
-> Exact sentence copy (2 highest-scored)
-> No new words generated
-> Preserves original phrasing exactly

"researchers found large black hole near
earth. dr chen at mit led the study using
webb telescope. mass 10 billion suns."
-> New words generated (abstractive)
-> Paraphrases: discovered->found
-> Sequential decode, one token at a time

"Scientists found a huge black hole close
to Earth. MIT team led by Dr Chen used
James Webb telescope to detect the object
with mass 10 billion times our Sun."
-> Best paraphrase quality (parallel attn)
-> discovered->found, massive->huge, near->close

>> Each summary independently enters all 4 evaluation modules ▼

reference
-> ROUGE
-> NER

STAGE 4 · EVALUATION

(1) N-GRAM OVERLAP [INTRINSIC]

WHAT IT MEASURES

Lexical overlap between generated summary and human-written reference summary.

ROUGE-1 (unigram)

$P = |S \cap R| / |S| \quad R = |S \cap R| / |R|$

$F1 = 2 \cdot P \cdot R / (P+R)$ [harmonic mean]

Example: S={scientists,found,huge,black,hole}

R={scientists,found,huge,black,hole,close}

$|S \cap R|=5$, $P=5/5=1.0$, $R=5/6=0.83$

$F1 = 2 \cdot 1.0 \cdot 0.83/1.83 = 0.907 = 90.7\%$

ROUGE-2 (bigram)

S-bigrams: (scientists,found)(found,huge)...

R-bigrams: (scientists,found)(found,huge)...

One synonym difference (planet vs Earth)

breaks ALL bigrams containing that word!

That is why ROUGE-2 is always << ROUGE-1

ROUGE-L (Longest Common Subsequence)

LCS(S,R) = longest common subsequence

(need not be contiguous)

$Precision_L = LCS / |S|$

$Recall_L = LCS / |R|$

$F_L = 2 \cdot P_L \cdot R_L / (P_L+R_L)$

Rewards preserving WORD ORDER

TF-IDF copies article order != ref order

-> TF-IDF ROUGE-L gap (+117%) > ROUGE-1 (+44%)

WHY HARMONIC MEAN?

Arithmetic: avg(1.0, 0.0) = 0.5 <- cheatable

Harmonic: $2 \cdot 1.0 \cdot 0.0/(1.0+0.0) = 0.0$ <- correct

Harmonic mean penalizes imbalanced P/R heavily.

CHEAT CASE 1 — Copy full article

Recall=1.0 Precision->0 F1->0 <- caught!

CHEAT CASE 2 — Output 1 correct word

(2) NER ENTITY PRESERVATION [INTRINSIC]

WHAT IT MEASURES

Are key named entities from the article preserved in the generated summary?

TOOLS USED

Primary: spaCy en_core_web_sm

Fallback: Rule-based regex patterns

Types: PERSON ORG GPE DATE NORP
FAC PRODUCT EVENT MONEY

HOW IT WORKS

1. Run NER on Article -> Article Entities (A)
2. Run NER on Summary -> Summary Entities (S)
3. Compute entity-level P, R, F1:

$P = |S \cap A| / |S|$ (no hallucinated ents)

$R = |S \cap A| / |A|$ (key ents preserved)

WORKED EXAMPLE

Article entities:

Sarah Chen	[PERSON]
MIT	[ORG]
Earth	[GPE]
James Webb	[FACILITY]

Transformer summary entities:

Dr Chen	[PERSON]	<- partial match
MIT	[ORG]	<- exact match
Earth	[GPE]	<- exact match

$|A|=4$, $|S \cap A|=3$ (Chen+MIT+Earth)

Precision = 3/3 = 100%

Recall = 3/4 = 75% (Webb missed)

KEY INSIGHT: CASING BUG

Transformer outputs lowercase tokens!

"earth" != "Earth" -> spaCy misses it

-> NER Recall = 0% for Transformer

Fix: capitalise first token, or use

(3) SENTIMENT CONSISTENCY [EXTRINSIC]

WHAT IT MEASURES

Does the summary preserve the SENTIMENT of the original article? (downstream task)

CLASSIFIER: LOGISTIC REGRESSION

Input features: TF-IDF vector (vocab ~5000)

Classes: POSITIVE NEUTRAL NEGATIVE

Training: labeled news articles

TF-IDF VECTORISATION

1. Tokenize text -> remove stop words
2. Build vocab (top 5000 by corpus TF-IDF)
3. Each doc -> sparse vector of TF-IDF scores

LOGISTIC REGRESSION — SOFTMAX

$z_k = w_k \cdot x + b_k$ for each class k

$P(k|x) = \exp(z_k) / \sum \exp(z_j)$

WORKED EXAMPLE

Article: "Scientists found a black hole..."

TF-IDF dominant: scientists=0.82

discovered=0.74, hole=0.71

$z_{pos} = w_{pos} \cdot x = 1.204$

$z_{neu} = w_{neu} \cdot x = 0.910$ <- WINS

$z_{neg} = w_{neg} \cdot x = 0.634$

softmax: neu=0.43 pos=0.34 neg=0.23

Predicted: NEUTRAL (science article)

CONSISTENCY CHECK

Article label: NEUTRAL

Summary label: NEUTRAL -> MATCH ✓

If summary=NEGATIVE -> MISMATCH ✗

Mismatch = summary distorts meaning

WHY EXTRINSIC?

Tests whether summary is useful for a DOWNSTREAM task (sentiment analysis),

(4) TOPIC CLASSIFICATION [EXTRINSIC]

WHAT IT MEASURES

Does the summary preserve the TOPIC of the article? (downstream classification)

CLASSIFIER: RANDOM FOREST

Trees: 30 decision trees

Features: TF-IDF (top ~1000 words)

Feature subset per split: sqrt(1000)=32

Bootstrap sampling per tree

Final: majority vote across 30 trees

TOPICS (10 classes)

Politics Sports Science Technology

Business Health Entertainment

World Arts Environment

DECISION TREE SPLIT (one tree, one node)

Parent S: {Sci:4, Tech:3, Pol:3} n=10

$H(S) = -(4/10 \cdot \log_2(4/10) + 3/10 \cdot \log_2(3/10)$

$+ 3/10 \cdot \log_2(3/10))$

$= 1.571$ bits [entropy]

Split on "telescope" > 0.3:

Left {Sci:3,Tech:1} Right {Tech:2,Pol:3}

$H(L) = 0.811 \quad H(R) = 0.971$

$IG = 1.571-(4/10 \cdot 0.811+6/10 \cdot 0.971)=0.395$

Choose split that maximises IG at each node

RANDOM FOREST ENSEMBLE

Tree 1 -> Science

Tree 2 -> Science

Tree 3 -> Technology

... (30 trees total)

Majority vote -> Science ★

EVALUATION

5-fold cross-validation: 86.2% +/- 1.8%

CONSISTENCY CHECK

STAGE 5 · METRICS & DASHBOARD

INTRINSIC vs EXTRINSIC EVALUATION — KEY DISTINCTION

INTRINSIC EVALUATION (Modules 1 + 2)

- Directly compares summary against reference / article text
- ROUGE: n-gram overlap with human reference summary
- NER: entity recall from original article
- Weakness: penalises valid paraphrases (synonym != word match)
- ROUGE misses: "found" vs "discovered" despite same meaning

-> Needs reference summary | Measures surface-level fidelity

EXTRINSIC EVALUATION (Modules 3 + 4)

- Tests whether summary is useful for a downstream NLP task
- Sentiment: does summary preserve positive/negative tone?
- Topic: does summary preserve the subject category?
- Strength: no reference needed, tests real-world usefulness
- Consistency mismatch = summary distorts meaning

-> No reference needed | Measures semantic faithfulness

METRICS AGGREGATION — PER-MODEL SCORECARD

ROUGE-1 F1:	17.0%	~20%	
ROUGE-2 F1:	6.1%	~8%	
ROUGE-L F1:	9.2%	~14%	
NER Precision:	100%	~90%	
NER Recall:	~60%	~65%	0%*
Sentiment Match:	✓ 78%	✓ 83%	
Topic Match:	✓ 81%	✓ 85%	

* Transformer NER Recall = 0%: lowercase token output defeats capitalization-based NER. Fix: capitalise first token or use neural NER.

STREAMLIT DASHBOARD — INTERACTIVE EVALUATION INTERFACE

- Bar chart: R1/R2/RL
- for TF-IDF / BiLSTM / Trans
- Side-by-side comparison
- Relative gain labels
- "+117% on ROUGE-L"
- Download as CSV

- Entity table per model
- PERSON ORG GPE FACILITY
- Highlight: missed entities
- Precision / Recall bar
- Casing bug warning flag
- Show article entities

- Pie chart: POS/NEU/NEG
- Article vs Summary
- Match indicator checkmark
- Softmax probability bars
- z-score per class shown
- Logistic Regression info

- Predicted topic: chip
- Confidence per class
- Decision tree visualised
- 30-tree vote histogram
- 5-fold CV score: 86.2%
- Feature importance plot

KEY INSIGHTS — INTERVIEW-READY CONCLUSIONS

> ROUGE-L gap (+117%) >> ROUGE-1 gap (+44%):

Transformer preserves compressed word ORDER matching reference; TF-IDF copies article order which diverges from reference structure.

> TF-IDF: High Precision / Low Recall:

Exact copies -> never fabricates words (high P), but cannot match paraphrased references (low R). Bounded by lexical overlap.

> Transformer paraphrases like humans:

"discovered"->"found", "massive"->"huge", "near"->"close to" — matches reference phrasing because humans also paraphrase. ROUGE rewards this.

> Extrinsic eval catches what ROUGE misses:

A summary with 0% ROUGE can still be semantically correct. Sentiment/topic consistency tests real-world downstream utility.